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Data Science (Assignment - 1)

1. Probability distribution -

1.1) A) Given;

$$P(X=x) = \{0.1, 0.4, 0.3, 0.2\} \text{ for } x \in \{1,2,3,4\}$$

for $(P(x \leq 3))$;

$$P(x \leq 3) = P(X=1) + P(X=2) + P(X=3)$$
$$= 0.1 + 0.4 + 0.3 = 0.8$$

$$E[X] = \sum x.P(x)$$
$$= 1.P(1) + 2.P(2) + 3.P(3) + 4.P(4)$$
$$= 1 \times 0.1 + 2 \times 0.4 + 3 \times 0.3 + 4 \times 0.2$$
$$= 0.1 + 0.8 + 0.9 + 0.8$$

$$Var(X) = E(X^2) - (E(X))^2$$
$$= (1^2 \times 0.1 + 2^2 \times 0.4 + 3^2 \times 0.3 + 4^2 \times 0.2) - (2.6)^2$$
$$= 7.6 - 6.76 = 0.84$$

B) Given, $\lambda = 3$ PMF :, $P(K=k) = \frac{\lambda^k e^{-\lambda}}{k!}$

a) $P(K=2) = \frac{3^2 e^{-3}}{2!} = \frac{9 \times 0.0498}{2} = 0.224$

b) $P(K \geq 1) = 1 - (P(K=0)) = 1 - \frac{3^0 e^{-3}}{0!} = 1 - 0.0428$

$= 0.950$ (Ans)

1.2) A) $f(y) = cy^2$; $0 \leq y \leq 2$

a) find c

$$\int_0^2 cy^2 dy = 1$$
$$\Rightarrow c\int_0^2 y^2 dy = 1$$
$$\Rightarrow c\left[\frac{y^3}{3}\right]_0^2 = 1$$
$$\Rightarrow c\left[\frac{8}{3} - 0\right] = 1$$
$$\Rightarrow c = \frac{3}{8}$$

b) $P(1 \leq y \leq 2) = \int_1^2 \frac{3}{8} y^2 dy = \frac{3}{8}\int_1^2 y^2 dy$

$$= \frac{3}{8}\left[\frac{y^3}{3}\right]_1^2 = \frac{3}{8}\left[\frac{8}{3} - \frac{1}{3}\right]$$
$$= \frac{\cancel{3}}{8} \times \frac{7}{\cancel{3}} = \frac{7}{8}$$

B) Given; $\lambda = 0.5$

PDF; $f(t) = \lambda e^{-\lambda t}$

a) CDF; $F(t) = \int_0^t 0.5 e^{-0.5u} du$

$F(t) = 1 - e^{-0.5t}$

b) $P(T>4) = 1 - F(4)$

$= 1 - (1 - e^{-0.5t}) = e^{-0.5 \times 4} = e^{-2} = 0.1353$

2) Hypothesis Testing:

Given; $n = 36$
$\bar{x} = 492$
$\sigma = 20$
$\mu_0 = 500$

1) Hypothesis
$H_0 : \mu = 500$
$H_0 : \mu < 500$ (left tailed test)

2) Z-statistics: $Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{492 - 500}{20/\sqrt{36}} = \frac{-8}{20/6}$
$= \frac{-8}{3.33} = -2.4$

3) Critical value:
for $\alpha = 0.05$
left tail $\Rightarrow Z_{0.05} = -1.645$

4) On comparing Z with $Z_{0.05}$;
$Z = -2.4 < -1.645$ left tailed.
$\Rightarrow$ Reject H_0; as it is left tailed.

3) P-value analysis: $Z = -2.4$ using $\alpha = 0.05$
same as 2. $Z_{0.05} = -1.645$

4) $P(Z < -2.4) = 0.0082$

5) $\because P(Z < -2.4)$ i.e $0.0082 < 0.05 (\alpha)$

$\boxed{\text{Reject } H_0}$ $\Rightarrow$ There is strong evidence the product weigh less than 500g.

4) PCA via SVD:

$D = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix}$

1) Column mean = Col 1 = $\frac{2+4+6}{3} = 4$
col 2 = $\frac{3+5+7}{3} = 5$

Centered matrix = $\begin{bmatrix} 2-4 & 3-5 \\ 4-4 & 5-5 \\ 6-4 & 7-5 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & -2 \end{bmatrix}$

2) Covariance matrix
$\Sigma = \frac{1}{n-1} Z^T Z$

$Z^T Z = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$

$\Sigma = \frac{1}{2}\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$

3) Eigen value; $|\Sigma - \lambda I| = 0$

$\begin{vmatrix} 4-\lambda & 4 \\ 4 & 4-\lambda \end{vmatrix} = 0$

$(4-\lambda)^2 = 16$

$\lambda(\lambda - 8) = 0$

$\boxed{\lambda = 8}$ $\boxed{\lambda = 0}$

Eigenvector:

for $\lambda = 8$; $\quad v = \frac{1}{\sqrt{2}}[1, 1]$

4) Interpretation:

$\because \lambda = 8 >>> \lambda = 0$, most variance lies along $(1, 1)$

Thus, first principal component captures maximum variance.

5) Data standardization:

Dataset: $S = \{12, 15, 14, 10, 18, 11, 18\}$

Mean; $\bar{x} = \frac{12+15+14+10+18+11+18}{7} = 13.29$

Standard, deviation, $s \approx 2.69$

Z-scores:

x	z
12	-0.48
15	0.64
14	0.26
10	-1.22
18	1.75
11	-0.85
13	-0.11

Reason, why standardization is necessary:

i) Features may have different units.

ii) Variables with larger scale dominate variance.

iii) PCA depends on variance magnitude.

iv) It ensures equal contribution from all features.

6) Foundation: Rank, Eigenvalue & Geometric Transformation

6.1) Rank & Nullity:

1) $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

2) Rank = no. of non-zero rows

= Rank(B) = 2

3) $\because$ rank $\neq n(B)$ $\Rightarrow$ Matrix is not invertible.

6.2) $A = \begin{bmatrix} 4 & 2 \\ 2 & 7 \end{bmatrix}$

1) Eigen values;

$|A - \lambda I| = 0$

$(4-\lambda)(7-\lambda) = 4$

$\lambda^2 - 11\lambda + 24 = 0$

$(\lambda - 8)(\lambda - 3) = 0$

$\lambda_1 = 8, \lambda_2 = 3$

2) Eigen vectors;

for $\lambda = 8$;

$v_1 = (1, 2)$

unit vector $= \frac{1}{\sqrt{5}}(1, 2)$

for $\lambda_2 = 3$;
$v = (2, -1)$
unit vector $= \frac{1}{\sqrt{5}}(2, -1)$

3) Orthogonality:

$(1, 2) \cdot (2, -1) = 2 - 2 = 0$

$\Rightarrow$ Matrix is orthogonal.

6.3) Geometric transformation:

1) Rotation (45°)

$$R_\theta = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & -\cos 45^\circ \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

2) point $= (1, 0)$

New point $= (\cos 45^\circ, \sin 45^\circ) = (0.707, 0.707)$

Translation:

Given,

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \qquad P = (2, 3)$$

$\Rightarrow$ Homogeneous co-ordinates $= (2, 3, 1)$

matrix; After translation: $t_x = 5, t_y = -2$

$$T = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiplying:; $P_\theta = (7, 1, 1)$

Final points $= (7, 1)$